



NATIONAL PRACTICAL EXAMINATIONS FOR TECHNICAL SECONDARY SCHOOLS (TSS), SCHOOL YEAR 2024-2025

SUMMATIVE/INTEGRATED ASSESSMENT

SECTOR: ICT & MULTIMEDIA TRADE: SOFTWARE PROGRAMMING AND EMBEDDED SYSTEMS (SPE)

Integrated Assessment Project: ...Apply Advanced Java Programming with OOP...

COMPETENCES ASSESSED

S/No	Code	Competence Names	RQF Level
1	SPEJO402	Fundamentals of OOP with Java	L4
2	SPEAD402	Advanced Database Development	L4
3	SPEAJ402	Advanced Java	L4
5	SPEAD402	Advanced Database development	L4

1. INTEGRATED SITUATION:

The Government of Rwanda employs thousands of employees across its different institutions. Each institution manages its finance, Human resources which are part of an Enterprise Resource Planning (ERP) system. Among the Functionalities expected from the ERP include the Employee Management and Payroll Management System(PMS).

Employee Management which is a Centralized storage of employee details including Personal Information and professional employment details.

The **Payroll Management System** is a core module of an enterprise resource planning (ERP) solution tailored for managing employee compensation by calculating the Employee Salary based institutions details, rules and regulations indicating taxes, deductions and base salary.

The Government wants to develop a new ERP system starting with the new updated rates of deductions to replace the existing system.

TASKS

As a Java developer, you have been hired to develop the Backend for Enterprise Resource Planning starting from the Payroll Management and Employee Management with the following minimum features:

Task 1: Employee Management

- Centralized storage of employee details, including:
 - Employee personal information: firstName, lastName, email, district, mobile, date of birth.
 - Employment or Professional information: Employee ID, department, position, salary and status(Active or Inactive), joining date.

It is important to note that salary added to employment details is a base Salary which helps to compute deductions, Gross salary and net salary.

Task 2: Perform Users Management with role based security. Implement the authentication and authorisation using JWT .

The user is an employee having username as email and password.

Users(id, Employee, password, status)

Users Roles to be implemented:

- ROLE_MANAGER role can process Salary and add Employee details
- ROLE_ADMIN can approve salary

- ROLE_EMPLOYEE can view his details, download payslip and can view pending salary payment

Task 3: Deductions and taxes Management

Generate endpoints allowing the management of deductions by keeping percentages of tax and deductions that should be deducted from the base salary.

The following deductions should be applied:

No	Deduction name	Percentage(%)
1	EmployeeTax	30
2	Pansion	6
3	MedicalInsurance	5
4	Others	5
5	House	14
6	Transport	14

Task 4: Design Payroll Management System database

- User
- Employee table
- Employment
- Deductions
- Payslip

Task 5: Payroll and payslip computation:

The Manager can start the payroll of a given Month/Year based on the salary basePay and deductions, the system should compute and generate the salary of all employees as a payroll. The employees should be active and you should make sure that all deductions don't exceed the Gross salary. Individual Employee can view his Payslip.

Example: Given the basicSalary of an Employee Peter as 700.000 FRW, the payslip should indicate

- Gross salary** =BaseSalary+House+Transport
 $=70.000+(70.000)*14/100+(70.000)*14/100=70.000+10.000+10.000$
=90.000
- netSalary**=baseSalary-(baseSalary*employeeTaxRate+baseSalary*medicalInsuranceRate+basePay*pansionRate+basePay*otherTaxRate).

Example of netSalary of Peter
$$=70.000-(70.000*30/100+70.000*6/100+70.000*5/100+70.000*4/100)$$
$$=70.000-(21.000+4200+3500+3500)$$
$$=57.800$$

Consider the following is a sample payslip of December 2024 payslip for RCA employees:

Emp id	Name	Base	House	Transport	Gross	Tax	Pension	Medical	Others	NetSalary	status	Month	Year
123	Mugabo Jarvis	70000	10000	10000	90000	21000	4200	3500	3500	57800	Pending	06	2025
234	Michou Michelle	35000	5000	5000	45000	10500	2100	1750	1750	28900	Pending		

Generate detailed payslips that employees can download or view. Include gross salary, deductions, taxes, and net pay details.

Task6: Configure Messaging using Database Routines.

When the ADMIN approves the payroll using the appropriate endpoint, configure at the DBMS level the Database routines of your choice(Procedure, Cursor and trigger). Generate a message at the database side that needs to be sent to the Employee in the following format when there is an insert operation to the payslip. Add an appropriate table message which indicates the Employee,message and Month-Year messages sent. Remember also to update the status of individual payslip to Paid.

Format of the message:

Dear <FIRSTNAME> Your <salary of MONTH/YEAR> from <INSTITUTION> <AMOUNT> has been credited to your <EMPLOYEE ID> account Successfully.

Instructions

- Prior to start coding, design the Entity Relationship Diagram(ERD) of the expected Database based on the relational DBMS of your choice.
- The backend must be designed using Spring boot. Spring Data JPA should be used to configure the database and generate APIs to be used for backend implementation.

- Design the Spring boot Flow Diagram indicating the functionality of the system.
- After proper configuration of the backend, the records of Users, Employee details, deductions and respective transactions should be added manually using either postman/swagger/application main class/DBMS level.
- Document your APIs using Swagger UI
- Perform authentication and authorisation using Json Web Token(JWT)
- Prevent duplicate payroll of a given employee in the same MONTH/YEAR.

Additional Information:

Time Allowed: 5 hours

Materials Provided: Computer